

Looking for knowledge management in library and information science research

Knowledge
management
in library

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Abstract

Purpose – The purpose of this paper is to investigate the place of knowledge management (KM) in library and information science (LIS) research. The authors looked for KM articles published in top LIS journals since 2000.

Design/methodology/approach – The authors analyzed publication trends and the distribution of articles by KM-related terms used, region/country and information-setting type. The authors also identified the top-cited KM articles published in these LIS journals.

Findings – In total, 672 KM-related articles (only 3.84%) were found within the 17,491 articles published in these LIS journals.

Research limitations/implications – This low percentage shows that while it might seem like LIS people are advocating for KM in libraries and other information settings, the authors find that despite the synergies, KM is yet to find a broader place within other LIS concerns.

Originality/value – Research published during the past two decades has frequently discussed how KM is important for libraries and for LIS. However, the authors do not know the extent to which KM forms a part of LIS research.

Keywords Bibliometrics, Publication trends, Google scholar journal ranking, Knowledge management research, Library and information science journals, Research publications

Paper type Research paper

Introduction and background

Knowledge management (KM) has gained much attention during the past three decades. The goal of KM is to gain competitive advantages by creating new knowledge and applying knowledge which creates value for organization (Alavi and Leidner, 2001). It involves “activities related to capturing, utilizing, creating, transferring, and storing of organizational knowledge” (Jasimuddin, 2005, p. 39). Although KM is highly practiced in the business sectors, the library and information science (LIS) community is also familiar with it. There have been many studies which show the relationship between KM and LIS (Sarrafzadeh, 2008; Roknuzzaman, 2009). Access to knowledge and application has always been a core concern for LIS professionals, and many think that KM is new wine in an old bottle (Rowley,



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1999; Schwarzwalder, 1999; Townley, 2001). KM in libraries can improve efficiency of library operations by managing the employee and user knowledge and helping meet user expectations (Wen, 2005). KM in nonprofit organizations such as libraries can improve communication among staff and between top managements and can promote a culture of sharing (Teng and Hawamdeh, 2002).

Many researchers in the past two decades have worked on the intersection of LIS and KM fields and discussed how KM is important for LIS, LIS professional associations (Agarwal and Islam, 2016) and for libraries. Some notable examples of research involving KM and academic libraries during the past two decades include building a knowledge bank in a university (Branin, 2003); reference services (Gandhi, 2004); KM from an African perspective (Jain, 2007); adoption of library 2.0 functionalities (Kim and Abbas, 2010); digital libraries (Islam and Ikeda, 2014); knowledge retention and transfer: how libraries manage employees leaving and joining (Agarwal and Islam, 2015); KM practices in Europe (Balagué *et al.*, 2016); KM readiness of academic libraries (Marouf, 2017); KM and service innovation (Islam *et al.*, 2017); Nigerian context (Ugwu and Ezema, 2018); and perspective of LIS professionals toward implementation (Ahmad, 2019). KM in LIS has also lead to doctoral dissertations in this area, e.g. implications of KM for LIS professions (Sarrafzadeh, 2008); the meeting of an existing and an emerging discipline (Roknuzzaman, 2009); e-learning in LIS schools from the viewpoint of KM (Islam, 2012); and KM for innovation in academic library services (Islam, 2016).

While there have been many studies conducted on KM in libraries and many more are currently being conducted, we do not know the extent to which KM forms a part of LIS research. In a previous study (Agarwal and Islam, 2018), the authors tried to ascertain the place of LIS in KM research. We found that only 1.8% of all articles in the top ten KM journals published since the year 2000 related to LIS.

In this paper, as opposed to looking for the place of LIS in KM research, our research objective is to investigate the place of KM in LIS research, i.e. the visibility of KM research in the LIS field. To fulfill this objective, we analyze the top 20 publications from 2000 to 2018 as listed by Google Scholar Metrics (2019) in the category LIS. To fulfill our research objective, we looked at a few research questions:

- RQ1. How many KM-related papers have been published in the top LIS journals since the year 2000? What is the percentage of KM articles among the overall LIS articles on different areas published in this period?
- RQ2. What are the publishing trends of these KM-related articles by year and by journal?
- RQ3. Which areas of KM research are gaining importance in these journals? In the KM-related articles in top journals during this period, what are the KM-related terms used; regions/countries investigated; information setting/library type investigated; and type of research methods used?
- RQ4. Which are the top-cited articles within the identified KM-related articles in top LIS journals during this period? What is the coverage of these top-cited articles in different journals and in different years? Which are the ten top-ranked articles based on number of citations and average citations per year?

Our analysis will help us determine the extent to which KM has been embraced by the LIS community. It also has other implications such as where LIS researchers writing about KM might want to publish for highest impact (in KM journals versus LIS journals), difference in publishing preferences between academics and practitioners based on competition versus sharing culture and the perceived versus actual synergies between the LIS and KM fields.

Literature review

Analyzing the research and publishing trends on a particular subject helps to identify the way a discipline grows and thrives (He *et al.*, 2012; White *et al.*, 2016). There have been many research studies conducted on the growth of KM literature and on other related aspects such as its geographical coverage or relationship with other disciplines. We discuss these here. Harman and Koohang (2005) used citation analysis to compare the frequency of publication and topical coverage of KM books and doctoral dissertations published from 1983 to 2005. They found that KM was emerging as a field of study and recommended KM topics for theorists and practitioners to focus on. Nonaka and Peltokorpi (2006) reviewed the top-20 most frequently cited KM articles in management journals. They classified these articles based on subjectivity–objectivity continuum and discussed their strengths and weaknesses. They also discussed the challenges of KM and how objectivity and subjectivity can be used in future research. Jasimuddin (2006) aimed to trace the root of KM and found that KM draws from a wide range of disciplines. Akhavan *et al.* (2016) reviewed KM literature from 1980 through 2014 and found that the most cited articles are from the USA and the UK. They found 2009 to be the prolific year, 2012 with the highest citations, and the *Journal of Knowledge Management* having the largest share in publishing the most cited articles in KM. Sedighi and Jalalimanesh (2017) mapped the research trends of KM from the Web of Science. They found that KM has changed and expanded dynamically during 2004–2010. In a bibliometric analysis of KM research, Gaviria-Marin *et al.* (2018) found that KM research is increasing, with the most publications in developed countries coming from the USA. In another bibliometric study on twenty years of the *Journal of Knowledge Management*, Gaviria-Marin *et al.* (2018) found that there is positive evolution of KM publications, with the USA and the UK leading the publications in this journal while emerging economies contributed fewer publications.

KM has been applied in multiple areas and in different disciplines Hallin and Marnburg (2008) reviewed empirical KM research in the hospitality field; Kothari *et al.* (2011) summarized the business literatures of KM which can be applied in the health sectors; Durst and Runar (2012) reviewed research on KM in small and medium-sized enterprises; Massaro *et al.* (2015) reviewed the public sector studies of KM; Xue (2017) reviewed publications of KM and its importance in organizations; Agarwal and Islam (2018) reviewed the extent to which LIS-related papers have been published in KM journals. None of these prior studies have investigated the extent to which KM research papers have been a part of LIS journals. In a very recent study, Ahmad *et al.* (2019) conducted a bibliometrics analysis of KM literature published in the field of LIS. They took data from the ISI Web of Science covering the period 1900–2017. They found that the USA has the highest number of research publications and 2007, 2012 and 2016–2017 were the most productive years in the area of KM in the field of LIS.

While the previously mentioned studies focus on the growth of KM and its application to different areas, including LIS, and while there have been studies related to KM in LIS journals, there is no study that has looked at the extent to which KM research forms a part of LIS research. Taking the opposite lens from our earlier study in which we sought to ascertain the place of LIS within KM research (Agarwal and Islam, 2018), in this study, we aim to determine the place of KM in LIS research. Examining the KM research areas in LIS journals would help both KM and LIS researchers and practitioners assess the extent to which LIS research has embraced KM-related research and provide directions for future steps.

Methodology

Our primary aim in this research was to determine the number of KM papers published in the 21st century in the top LIS journals. There are different databases available to retrieve

top journals, such as Web of Science, Scopus and Google Scholar Metrics, all with their relative pros and cons (Yang and Meho, 2006; Martin-Martin *et al.*, 2019; Iowa State University Library, 2020). In recent years, Google Scholar has emerged as a serious alternative to Scopus and Web of Science (Harzing, 2019; Google Scholar Metrics, 2019). Google Scholar has better interdisciplinary coverage compared to Web of Science, includes more non-English language publications compared to Scopus and Web of Science (Iowa State University Library, 2020), and it addresses the limited coverage that Web of Science listing of journals in social science and humanities compared to its sciences listing (Harzing, 2019). Harzing concludes how Google Scholar has democratized citation analysis. "Rather than leaving it in the hands of those with access to commercial databases with high subscription fees, everyone with a computer and internet access can now run their own analyses" (Harzing, 2019).

In most past studies examining the availability of literature of KM in LIS, data was imported from the Web of Science or Scopus databases Ahmad *et al.*'s (2019) study relating KM and LIS; changing trends in KM research from an LIS perspective (Ahmadi and Nazim, 2017); and an analysis of global research productivity in KM using the Scopus database (Khiste *et al.*, 2018). While past bibliometrics studies in the LIS and KM space have used Web of Science and Scopus databases, there are no studies examining LIS and KM that used the journal listing from Google Scholar, a free accessible Web search engine that indexes the full text or metadata of scholarly literatures.

For all these reasons, we decided to use Google Scholar Metrics (2019) in our study, which are derived from Google Scholar. To determine the top journals, we used the list of the top 20 publications matching LIS ranked by Google Scholar Metrics (2019) as on January 6, 2019. The ranking can change over time. Of these, we excluded those that pertained to conference proceedings (ranked 12, 18) and were not journals. We excluded one journal (arXiv Digital Libraries [cs.DL] ranked 3), as it was specifically focused on digital libraries in different domains, including machine learning and physics. This left us with a list of top 17 peer-reviewed journals among the 20 top publications identified by Google Scholar (Table 1). These are all peer-reviewed publications dedicated to the exchange of the latest research and practice in LIS. *Journal of the Association for Information Science and Technology (JASIST)* is the top-ranked in the LIS field with a 2019 Impact Factor of 2.835 (JASIST, 2019). Table 1 lists the ranking, as well as the h5-index and h5-median (productivity and citation metrics developed by Google Scholar) of these journals. H5-index is the h-index for articles published in the past five complete years. It is the largest number h such that h articles published in the past five years have at least h citations each (Journal-level impact, 2019). Thus, e.g. for *JASIST*, the h5-index is 53. H5-median is based on h5-index but instead measures the median (or middle) value of citations for the h number of citations (Journal-level impact, 2019).

Table 2 provides more information on the top-17 LIS journals. It lists the publisher, country, its frequency of publishing and the reputed databases indexing the journal. EBSCO (Elton B. Stephens Co), SCOPUS, Social Science Citation Index (SSCI), ProQuest, ISI Web of Science (WOS), Computer Science Index (CSI), Book Review Index (BRI) and Directory of Open Access Journals (DOAJ), are leading providers that index and abstract journals.

Keywords and search terms for KM-related articles can be based on literature or expert reviews. As authors, we are familiar with the KM literature and KM-related terms, as we have been conducting research in this field for more than a decade on average between us, are teaching the course in our relative universities and have used research-based KM textbooks such as Dalkir (2017). From the journals identified, we accessed the individual issues of each journal from the years 2000 – 2018. For each issue, we scanned the article title

Rank	Journal name	H5-index	H5-median	Remarks
1	<i>Journal of the Association for Information Science and Technology (JASIST)</i>	53	89	—
2	<i>Scientometrics</i>	49	63	—
3	<i>arXiv Digital Libraries (cs.DL)</i>	48	73	Excluded
4	<i>Journal of Informetrics (JOI)</i>	36	50	—
5	<i>Online Information Review (OIR)</i>	26	41	—
6	<i>College and Research Libraries (C&RL)</i>	25	39	—
7	<i>Library and Information Science Research (LISR)</i>	25	39	—
8	<i>The Journal of Academic Librarianship (JAL)</i>	25	35	—
9	<i>Journal of Documentation (JD)</i>	22	36	—
10	<i>Journal of Information Science (JIS)</i>	22	28	—
11	<i>Library Hi Tech (LHT)</i>	20	36	—
12	<i>International ACM/IEEE Joint Conference on Digital Libraries</i>	20	31	Excluded
13	<i>Reference Services Review (RSR)</i>	20	28	—
14	<i>The Electronic Library (TEL)</i>	20	28	—
15	<i>Portal: Libraries and the Academy (PLA)</i>	20	25	—
16	<i>Aslib Journal of Information Management (AJIM)</i>	20	24	—
17	<i>Journal of the Medical Library Association (JMLA)</i>	19	31	—
18	<i>Proceedings of the Association for Information Science and Technology</i>	19	25	Excluded
19	<i>Journal of Librarianship and Information Science (JOLIS)</i>	19	24	—
20	<i>Information and Learning Science (ILS) New Library World (NLW)</i>	18	35	—

Table 1.
Top 20 LIS journals
included in the study
ranked by Google
Scholar (Google
Scholar, 2019)

Table 2.
Rank, publisher,
frequency and
indexing database
for top 17 LIS
journals

Rank	Journal name	Publisher (country)	Frequency	Indexed by (impact factor)
1	<i>JASIST</i>	Wiley-Blackwell (USA)	Monthly	ProQuest, EBSCO, SCOPUS (2019: 2.835)
2	<i>Scientometrics</i>	Springer (Germany) and Akadémiai Kiadó (Hungary)	Monthly	SCOPUS, EBSCO, ProQuest (1979–2019: 2.173)
4	<i>JOI</i>	Elsevier (The Netherlands)	Bi-monthly	Scopus, WOS, EBSCOhost, SSCI, PubMed (2017: 3.484)
5	<i>OIR</i>	Emerald (UK)	Bi-monthly	Proquest, Scopus, ERIC, CompuMath Citation, Discovery, SSCI
6	<i>C&RL</i>	Association of College and Research Libraries (USA)	Bi-monthly	Scopus, SSCI, PASCAL, EBSCO(2017: 1.626)
7	<i>LISR</i>	Elsevier (The Netherlands)	Quarterly	Scopus, SSCI, Scirus, (2017: 1.372)
8	<i>JAL</i>	Elsevier (The Netherlands)	Bi-monthly	Education Index, Scopus, SSCI (2017: 2.32)
9	<i>JD</i>	Emerald (UK)	Bi-monthly	ProQuest, SSCI, Scopus, EBSCO (2017: 1.157)
10	<i>JIS</i>	Sage (UK)	Bi-monthly	Scopus, WOS, Social and Behavioral Sciences, Inspec, SSCI (2017: 1.939)
11	<i>LHT</i>	Emerald (UK)	Quarterly	BRI, CSI, SSCI, Scopus, (2017: 0.759)
13	<i>RSR</i>	Emerald (UK)	Quarterly	BRI, Zetoc, Fordham Francis Index (FRANCIS)
14	<i>TEL</i>	Emerald (UK)	Bi-monthly	CSI, SSCI, Scopus (2017: 0.800)
15	<i>PLA</i>	Johns Hopkins U. Press, USA	Quarterly	SSCI, WOS, EBSCO, ProQuest, (2017: 1.473)
16	<i>AJIM</i>	Emerald (UK)	Bi-monthly	SSCI, EBSCO, ProQuest, (2017 Impact Factor: 1.461)
17	<i>JMLA</i>	Univ. of Pittsburgh, USA	Quarterly	DOAJ, Cite factor(2017: 1.392)
19	<i>JOLIS</i>	Sage (UK)	Quarterly	SSCI, WoS, ERIC databases (2017: 1.089)
20	<i>ILS</i>	Emerald (UK)	Bi-monthly	Scopus, SSCI

looking for any relationship with “knowledge management” (e.g. KM, knowledge creation, knowledge sharing, big data, taxonomy, knowledge diffusion, data mining, knowledge culture, knowledge repository and KM readiness) or any relationship with KM even if the term was not specifically mentioned. For further clarification to look for this relationship, we read the abstracts of specific articles. Also see KM-related terms identified in the LIS journal articles from 2000 to 2018 (frequency).

From each KM-related article identified, we entered its bibliographic information, number of times it was cited (as per Google Scholar), its objectives and research questions, findings, type of research method used, region, type of information setting for the study and main keywords from the study. We compiled all this information in an Excel spreadsheet for all articles that we identified. We carried out the work of journal selection, combing of all articles and analysis of identified KM-related articles over a three-month period from January to March, 2019.

Data analysis and findings

When analyzing the gathered data from the 672 KM-related articles from top seventeen LIS journals over an 18-year period, we found a number of interesting trends addressing the four research questions. We discuss the findings for each question below.

RQ1. [Table 3](#) shows the number of KM-related articles published since 2000 in the top 17 LIS journals. The years are arranged starting from the most recent. Each row shows the total number of articles published by the LIS journal per year and the total number of KM-related articles (in brackets) published by that journal that year. The shaded column (second from left) shows the total number of articles published by each journal since 2000 and the LIS-related articles published by that journal during the time. The shaded row at the bottom shows the total number of KM articles published each year across the top 17 journals and the LIS-related articles published that year across all those journals. In total, we found that of the 17,491 articles published by the top 17 LIS journals from 2000 to 2018, 672 were KM-related.

[Table 4](#) shows the percentage of KM-related articles published among all LIS articles in top LIS journals since the year 2000. Each cell of the table represents a particular journal in a given year. Only those cells where KM-related articles were published in a certain year for that journal are populated. The values in those cells represent the percentage of KM-related articles published that year in that journal with respect to the total number of LIS articles published that year by that journal. For example, in 2018, 4.50% of the articles published in the JASIST were KM-related. In total, since 2000, 3.84% of the LIS articles were KM related.

RQ2. The last row in [Table 4](#) shows the percentage of KM-related articles published per year across the top LIS journals ([Figure 1](#)). From the figure, we can see that the year 2017 was the most productive in terms of the place of KM in LIS research, with 4.70% of all KM articles published across journals being LIS-related. From [Table 3](#), we can see that this contribution was by Scientometrics, when it accepted 347 LIS-articles for publication that year out of its total of 19 KM articles that it published that year.

[Figure 2](#) shows the publishing trends of KM-related articles in LIS journals per journal during the period from 2000 to 2018. The first shaded column in [Table 4](#) shows the percentage of KM-related articles published by each journal from 2000 to 2018. From [Figure 2](#), we can see that *Journal of Information Science* published the maximum number of articles in the KM area (8.36%) during the 18-year period from 2000 to 2018. However, in the top LIS journal *JASIST*, only 4.84% of the articles were KM-related during the same period.

RQ3. KM-related terms identified in the LIS journal articles from 2000 to 2018 (frequency) shows the KM-related terms found in the 17 LIS journal article titles and/or abstracts from

Table 3.
Total number of
articles publisher per
year per journal and
KM related articles
within those
(2000–2018)

Jn. Y.	Total	2018	2017	2016	2015	2014	2013	2012	2011	2010	2009	2008	2007	2006	2005	2004	2003	2002	2001	2000
JASIST	2,977(144)	111(6)	190(11)	214(7)	174(12)	180(14)	180(12)	176(7)	183(9)	178(7)	206(8)	184(5)	187(13)	167(5)	115(8)	105(6)	106(4)	107(6)	107(2)	107(3)
Scientia	3,561(171)	332(21)	347(19)	271(16)	258(7)	294(15)	253(18)	182(8)	218(11)	227(12)	191(7)	121(4)	131(5)	158(8)	110(3)	102(7)	91(4)	84(4)	110(1)	83(1)
JOI	884(26)	84(6)	110(6)	94(0)	83(1)	90(3)	122(2)	78(0)	63(0)	63(0)	32(6)	33(0)	32(1)	—	—	—	—	—	—	—
OIR	939(57)	70(5)	63(5)	62(1)	52(6)	51(2)	48(2)	49(1)	49(4)	51(8)	66(12)	56(3)	53(0)	46(1)	43(3)	41(0)	35(1)	32(1)	33(2)	39(0)
C&EL	691(11)	48(1)	49(3)	43(0)	57(0)	41(1)	33(0)	30(0)	30(1)	33(0)	32(1)	32(0)	33(1)	34(0)	30(0)	28(0)	28(1)	38(0)	35(2)	37(0)
LISR	538(13)	31(1)	34(0)	38(0)	41(1)	24(1)	36(1)	36(0)	37(2)	29(2)	27(0)	30(1)	25(1)	28(0)	26(1)	25(0)	21(1)	17(0)	17(1)	16(0)
JAL	1,193(18)	94(3)	60(0)	86(0)	101(3)	88(3)	77(2)	50(0)	65(0)	61(4)	57(0)	58(0)	76(0)	69(0)	69(1)	50(2)	36(0)	33(0)	29(0)	33(0)
JD	843(55)	71(2)	69(7)	60(4)	63(6)	55(2)	43(4)	43(5)	43(4)	42(3)	42(1)	43(1)	41(3)	35(5)	43(0)	33(1)	28(0)	26(3)	32(2)	31(2)
JIS	969(81)	52(2)	53(2)	58(4)	59(3)	65(4)	60(5)	41(2)	51(1)	51(6)	47(4)	55(7)	49(10)	45(6)	44(4)	46(7)	53(2)	52(4)	41(5)	47(3)
LHT	888(15)	43(4)	44(1)	46(0)	36(0)	46(0)	46(0)	48(1)	53(2)	47(0)	48(1)	51(1)	49(1)	51(0)	49(1)	47(1)	50(0)	51(1)	52(1)	31(0)
RSR	753(8)	41(0)	33(1)	34(1)	40(0)	41(0)	41(0)	39(0)	39(0)	45(0)	38(0)	36(1)	42(0)	50(0)	43(0)	42(2)	43(1)	38(0)	34(2)	34(0)
TEL	1,059(34)	72(2)	72(3)	60(3)	70(1)	52(1)	49(3)	50(3)	51(0)	57(0)	71(1)	61(2)	55(0)	61(6)	63(1)	55(1)	54(0)	42(4)	33(0)	31(3)
PLA	629(0)	35(0)	48(0)	40(0)	41(0)	39(0)	24(0)	23(0)	28(0)	25(0)	34(0)	29(0)	34(0)	34(0)	37(0)	33(0)	45(0)	41(0)	41(0)	—
AJIM	194(16)	38(3)	45(3)	40(1)	36(6)	35(3)	—	—	—	—	—	—	—	—	—	—	—	—	—	—
JMLA	124(0)	34(0)	41(0)	49(0)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
JOLIS	338(12)	16(0)	34(2)	28(2)	28(0)	24(0)	23(0)	23(1)	21(0)	22(2)	14(0)	20(0)	19(1)	17(1)	17(1)	15(0)	17(1)	19(1)	21(0)	20(0)
ILS	851(11)	60(2)	47(0)	52(0)	52(1)	42(0)	48(0)	48(1)	48(0)	49(0)	48(0)	48(0)	48(0)	42(0)	42(2)	37(1)	37(1)	37(3)	42(0)	24(1)
Total per year	17,491(672)	1,232(67)	1,339(63)	1,275(69)	1,191(47)	1,168(49)	1,083(49)	916(28)	975(35)	980(44)	953(41)	857(25)	874(36)	837(32)	731(25)	659(28)	644(16)	617(27)	627(18)	533(13)

Jn. Y.	% per Jour.	2017	2016	2015	2014	2013	2012	2011	2010	2009	2008	2007	2006	2005	2004	2003	2002	2001	2000
JASIST	4.84	450	579	327	690	778	667	398	492	393	388	272	695	299	696	571	377	561	187
Sciento.	4.80	633	548	590	271	510	711	440	509	529	366	331	382	506	273	686	440	476	280
JOI	2.94	714	545	000	120	333	164	000	159	000	1875	000	313	-	-	-	-	-	120
OIR	6.07	794	794	161	1154	392	417	204	816	1569	1818	536	000	217	698	000	286	313	606
C&RL	1.59	208	612	000	000	244	000	000	333	000	313	000	303	000	000	000	357	000	571
LISR	2.42	323	000	000	244	417	278	000	541	690	000	333	400	000	385	000	476	000	000
JAL	1.51	319	000	000	297	337	260	000	000	656	000	000	000	000	145	400	000	000	000
JD	6.52	282	1014	667	952	364	930	1163	930	714	238	233	732	1429	000	303	000	1154	625
JIS	8.36	385	377	690	508	615	833	488	196	1176	851	1273	2041	1333	909	1522	377	769	638
LHT	1.69	930	227	000	000	000	000	208	377	000	208	196	204	000	204	213	000	196	192
RSR	1.06	000	303	294	000	000	000	000	000	000	000	278	000	000	000	476	233	000	588
TEL	3.21	278	417	500	143	192	612	600	000	000	141	328	000	984	159	182	000	952	000
PLA	0.00	000	000	000	000	000	000	000	000	000	000	000	000	000	000	000	000	000	968
AJIM	8.25	789	667	250	1667	857	-	-	-	-	-	-	-	-	-	-	-	-	-
JMLA	0.00	000	000	000	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
JOLIS	3.02	000	588	714	000	000	000	435	000	909	000	000	526	588	588	000	588	526	000
IIS	1.29	333	000	000	192	000	000	000	000	000	000	000	000	000	476	270	811	000	417
% per year	3.84	462	470	305	394	419	452	305	358	448	430	291	411	382	341	424	437	287	243

Table 4.
Percentage of KM
related articles
among LIS articles in
top 17 LIS journals
(2000–2018)

Figure 1.
Percentage of KM
related articles
(among all LIS
articles) per year
across the top-17 LIS
journals

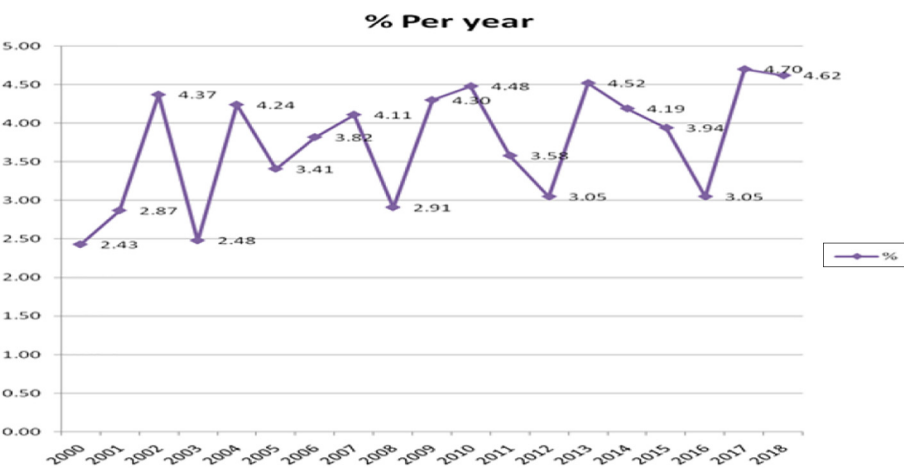
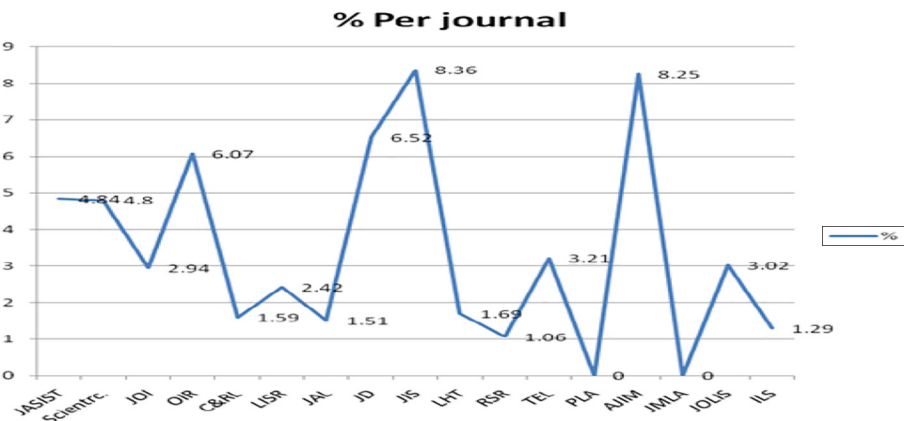


Figure 2.
Percentage of KM
related articles
(among all LIS
articles) per journal
for 2000–2018



2000 to 2018. Where there were related terms of a particular category, the frequency of these is listed in brackets in the first column of the table. Of the 672 unique KM-related articles identified, the total frequency of LIS-related terms was 674 (a particular article was found to touch-upon 1–3 terms listed in the table):

KM terms appearing in LIS journal articles (674):

- *Knowledge management* (249). knowledge sharing/dissemination/ knowledge spillover/transfer (75); KM (46); scientometrics/bibliometrics (32); knowledge creation (16); information literacy (14);); KM implementation/application (14); interdisciplinary collaboration/collaboration (13); KM culture (13); personal KM (12); knowledge acquisition (9); knowledge retaining (1); knowledge power (1); knowledge trading (1); knowledge extraction (1); and KM challenges (1).
- *Knowledge organization* (147). knowledge map/knowledge construction (29); organizational learning (26); taxonomy/knowledge classification (22); KM system (17); text mining/ content analysis/citation analysis/network analysis (14); open/big data (14); KM

- Ontology (8); knowledge tree (4); knowledge hub/knowledge space/knowledge commons (4); KM model (3); KM repository (2); terminology register (1); knowledge society (1); vocabulary development (1); and knowledge communication (1).
- *Knowledge types (108)*. knowledge diffusion (29); tacit knowledge (12); cognitive science (12) knowledge domain (11); KM in digital library (9); explicit knowledge (8); community practice/virtual communities (7); indigenous knowledge (5); knowledge development (2); KM-based economy (2); lexical knowledge (2); knowledge engineering (2); meta knowledge (1); scientific knowledge (1); knowledge resource (1); crowd sourced knowledge (1); Primary knowledge (1); procedural knowledge (1); and local knowledge (1).
 - *Technology/social networking tools (78)*. complex networks/knowledge network/gatekeeper/citation network (28); internet (14); ICT/e-tools (11); Wikis (9); Web 2.0(5); social media/technology (5); knowledge discovery tools (4); Twitter (1); and digital knowledge (1).
 - *Knowledge retrieval(52)*. semantic web (16); knowledge flow (10); knowledge production (5); knowledge retrieval (5); knowledge seeking (5); knowledge integration (5); knowledge searching (3); knowledge source (2); and knowledge gap (1).
 - *KM education (19)*. global KM education (9); and KM courses (10).
 - *KM innovation (15)*. business innovation (13); and innovation in system (2).
 - *User (8)*. knowledge workers (6); knowledge processionalists (1); and KM librarian (1).

Table 5 lists the distribution of KM articles in LIS journals by regions/countries, by information setting/library type and by the type of research method used. Most articles are from the USA, followed by China, Taiwan and the UK. The largest number of articles (417 of 672) pertained to general/other information settings which are followed by academic library. From the table, we can see that majority of KM studies in LIS journals were either reviews of the literature, opinion papers or theoretical papers. In total, 125 of the 472 studies described surveys that were carried out.

RQ4. Table 6 lists the top-cited KM-related articles published in LIS journals since 2000. Only top 20 articles were included (out of the 672 articles identified). Column 2 shows the number of citations (as per Google Scholar) received for each article. Column 3 shows the ranking of these articles (from 1 to 20) based on the total number of citations received (the top five ranks are shaded). However, some articles received citations in many years,

Region/country	Region/country not applicable (35); USA(128); China (74); Taiwan (64); The UK (48); South Korea (35); Spain (29); Canada (22); Australia (17); Finland (15); Singapore and Germany (13); New Zealand, Iran, Hong Kong and Brazil (11); Sweden, South Africa and Netherlands (10); Italy and France (9); Israel and Denmark (8); Japan (7); Mexico (6); Belgium (5); Poland and Nigeria, Qatar, Pakistan, India, Greece and Columbia (4); Turkey, Slovenia, Russia, Portugal, Kuwait, Hungary, Croatia, Argentina (2); and United Arab Emirates, Tanzania, Switzerland, Sierra Leone, Scotland, Saudi Arabia, Romania, Philippine, Malaysia, Lithuania, Kenya, Ireland and Botswana for each (1)
Setting type	Academic library (193); special library (53); public library (09); other information setting/general(417)
Method type	lit. review/opinion/theoretical (470); survey (125); interviews/case study (69); mixed methods (5); observation (3)

Table 5.
KM articles in LIS
journals –
distribution by
region/country,
setting type and
research method type
(N = 672)

Table 6.
Top 20 KM article
citations (from across
top 17 LIS journals,
2000–2018)

Overall ranking (1)	No. of citations (2)	Rank based on no. of citations (3)	Year since publication (4)	Average citations per year (5)	Rank based on average citations per year (6)
1 Lin (2007)	1,202	1	12	100.17	1
2 Zins (2007)	725	2	12	60.42	2
3 Alavi and Tiwana (2002)	648	3	17	38.12	4
4 Liao <i>et al.</i> (2007)	546	4	12	45.50	3
5 McJorney (2002)	521	5	17	30.65	6
6 Frické (2009)	334	8	10	33.40	5
7 Wildermuth (2004)	355	7	15	23.67	12
8 Hall (2001)	383	6	18	21.28	15
9 Baptista Nunes <i>et al.</i> (2006)	314	10	13	24.15	11
10 Garcia-Penalvo <i>et al.</i> (2010)	271	15	09	30.11	7
11 Widén-Wulff and Ginman (2004)	326	9	15	21.73	14
11 Yang and Chen (2007)	298	13	12	24.83	10
11 Seonghee and Boryung (2008)	277	14	11	25.18	9
14 Kim and Abbas (2010)	233	19	09	25.89	8
15 Blair (2002)	311	11	17	18.29	17
15 Holsapple and Joshi (2004)	298	12	15	19.87	16
17 Razmerita <i>et al.</i> (2009)	235	17	10	23.50	13
18 Townley (2001)	267	16	18	14.83	19
19 Garfield (2004)	235	18	15	15.67	18
20 Lim and Klobas (2000)	216	20	19	11.37	20

while others were highly cited within a shorter period. Column 4, thus, lists the year since publication (2019 – the year of publication). Column 5 shows the average citations received by each article per year (no. of citations/years since publication). The last column shows the rank received by the article (from 1 to 20) based on average citations per year (the top five ranks are shaded). An overall ranking was calculated based on the average of these two rankings (Columns 3 and 6). This was used to arrive at the list of the most influential KM articles in LIS journals (2000–2018). The number in the first column shows this ranking, and articles are arranged in order of influence, with the most influential articles listed on top. A few ranks are repeated when they had the same overall ranking. The top five most influential articles are shaded.

The 20 articles listed in Table 6 were analyzed for the journals in which they were published and the years in which these articles were published (Table 7). We found that *Journal of Information Science* published most of these top-cited KM articles.

Of these 20 top-cited articles, four were published in 2004 and 2007 and three in 2002.

In Table 6, we identified the five top-ranked KM articles in LIS journals published between 2000 and 2018 based on the number of citations and average citations per year. In Table 8, further details on these top articles are listed – the region or country where the study population was drawn from, the information setting or library type for the particular research, primary stakeholder that the study focuses on, the methodology used in the study and keywords describing the study.

Co-authorship patterns address the collaboration trends between the authors. To provide a comprehensive picture of KM research collaborations, we have examined the co-authorship patterns of KM articles in LIS journals. Table 9 shows the overall trends of authorship patterns of KM-related articles during 2000–2018. Among the 672 KM articles, majority of the publications (36.46%) were written by double authors, and the next largest was written by single authored (29.32%). We found a declining trend in multiple authorship works over time. It is evident that the number of studies published by multiple authors is significantly lower than the number of single-authored publications.

Discussion and implications

The aim of this study was to ascertain the place of KM in LIS research. To determine this, we looked for KM research published within the top LIS journals since 2000. We found that less than 4% of all research in LIS is KM-related (672 out of 17,491). In a yearly analysis, the percentage of KM research in LIS journals went up and down since 2000 and went up to 4.7% in 2017. A reason could be that KM researchers in LIS are publishing more in KM journals and less in LIS journals – where it might get greater visibility. KM-researchers in LIS might be wanting to get to their primary KM audience for more practical usage and citation impact, as opposed to publishing in LIS journals, which other KM researchers may not read as much. A recent study validates our finding of low occurrence of KM within LIS journals. A bibliometric analysis of 500 most-cited LIS articles from 1980 to 2017 showed that academic libraries, information literacy were the most popular keywords (with more

Table 7.
Coverage of top-cited
articles – in different
journals and in
different years
(*n* = 20)

Name of journals (no. of top-cited articles)	
Journals (no. of articles)	JIS (7); JASIST(6); OIR (2); LISR(1); C&RL(1); JAL(1); TEL(1); JD(1)
Years (no. of articles)	2004(4); 2007(4); 2002(3); 2010(2); 2009(2); 2001(2); 2008(1); 2006(1); 2000(1)

Table 8.
Five top-ranked KM-
related articles in LIS
journals (2000–2018)

Rank	Article	Region/country	Information setting/type	Primary stakeholder	Method	Keywords
1	Lin (2007)	Taiwan	General	Employees	Survey	Knowledge sharing, extrinsic motivation; intrinsic motivation
2	Zins (2007)	Global	General	Experts	discussion	Data, information, knowledge
3	Alavi and Tiwana (2002)	Not applicable	General	Experts	Theoretical	KM, knowledge integration
3	Liao <i>et al.</i> (2007)	Taiwan	Firms	Employees	Survey	KM, knowledge sharing, innovation
5	McInerney (2002)	Not applicable	General	Employees	Theory	KM, knowledge sharing, knowledge transfer

than 20 occurrences), followed by education and open access (more than 12 occurrences). Terms such as information retrieval, KM, information management, data mining, collection development and classification were less used, with 0–5 occurrences (Hodonu-Wusu and Lazarus, 2018). Also, for KM researchers working in LIS, the motivation to publish in KM versus LIS journals (and specific journals within these) might differ based on whether the researcher is an academic or a practitioner. An academic may be guided by more competition related to tenure and promotion, where as a librarian might be guided by sharing and practical usage.

Among the leading LIS journals, the highest number of KM-related articles was published in the *Journal of Information Science (JIS)* and the *ASLIB Journal of Information and Knowledge Management (AJIKM)*. Both these journals published more KM-related articles compared to other LIS journals. Among the KM-related articles, a majority focused on key terms within the general knowledge management discipline (the second row in KM-related terms identified in the LIS journal articles from 2000 to 2018 [frequency]) as their content area. This might be because of the limited inroads that LIS has made within KM, whereby a lot of authors still focus on the more popular terms/areas within KM. Most identified articles were theoretical and position papers. Limited work has been done on practical implementation of KM within libraries. As a lot of published research is still about advocating KM, a lot of the articles are still theoretical and position papers. In the KM-related articles, the research was mostly from the USA, China, Taiwan and the UK. It is in line with the Haimila's (2018) study which found that the top-listed 100 companies that matters in KM headquartered in the US and the UK. China and Taiwan, as fast emerging economies, also seem to be advocating KM within libraries.

In terms of citation impact, JIS and JASIST are good avenues for KM-researchers in the LIS area, as these two journals had most of the top-cited articles. Looking at the areas of focus in top five most-cited articles shows that KM, knowledge sharing, knowledge integration, knowledge transfer and innovation are major areas of interest to other researchers citing KM-in-LIS-related work. These are all important areas when implementing KM within libraries and other information settings. The overall findings of this study revealed that KM has comparatively high visibility (3.84%) in the LIS field, as opposed to LIS within the KM field, where Agarwal and Islam's (2018) study demonstrated a visibility of only 1.8%. However, as such, less than 4% is still a low figure. The low percentage of KM articles within LIS articles is owing to the overall image of KM as something for businesses and for for-profit companies.

Limitations and future works

The present study has a few limitations. First, we determined the top LIS journals and publications based solely on Google Scholar (as on January 6, 2019). There were other scholarly databases, e.g. SSCI, EBSCO, SCOPUS and Web of Science, which might provide

Table 9.
Pattern of single-
authorship versus co-
authorship in KM-
related publications
within LIS journals
(2000–2018)

No. of authors	Journal articles	Total publications (%)
Single-authored	197	29.32
By two authors	245	36.46
By three authors	122	18.15
By four authors	70	10.42
By five authors	23	3.42
By six or more authors	15	2.23
Total	672 articles	100

different listings of top LIS publications. Even within Google Scholar, the determination of top publications in a field can change over a few months. Some scholarly databases categorize publications based on tier (tier1, tier2. . .), quality (q1, q2. . . .) and/or impact factors. Future studies can consider these in the selection of top publications. Second, the research showed that despite the perceived synergies between the LIS and KM fields, KM research has had little impact within LIS publications. A qualitative study could be carried out to further delve into the reasons behind this scenario. Third, some of the journals classified under LIS, e.g. *JASIST*, are interdisciplinary and more information science-focused as opposed to library-focused. Here, not just KM, but library-specific research might also be small in number. Future research should look at co-occurrences of library-specific terms and KM terms in specific journal papers in the LIS list. Fourth, there is a bigger question to examine. While we seek to determine the place of KM within LIS research, the question still remains as to whether KM needs to have a place within LIS research and what, if any, are the synergies within and between the two fields. Finally, there is considerable scope for cross-disciplinary research here, given the involvement of KM in a wide range of subject areas. Although the focus of this paper specifies LIS, the methods and outcomes of the paper could direct the way forward for other disciplines that are part of the larger KM umbrella to investigate the place of KM in research within each of those fields.

We set out to determine the place of one field within another and arrived at an interesting set of findings. These would be of interest to LIS researchers and professionals in the field and those related to the KM profession. By finding that KM-related research in LIS is less than 4%, some LIS and KM professionals and researchers could argue that there is no point in KM trying to be visible within LIS research, as the area is too far removed to make a huge difference. On a more positive note, as various researchers have already argued about the value of KM for libraries, we call upon both LIS and KM researchers and practitioners to do more to be recognized as important areas to each other.

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